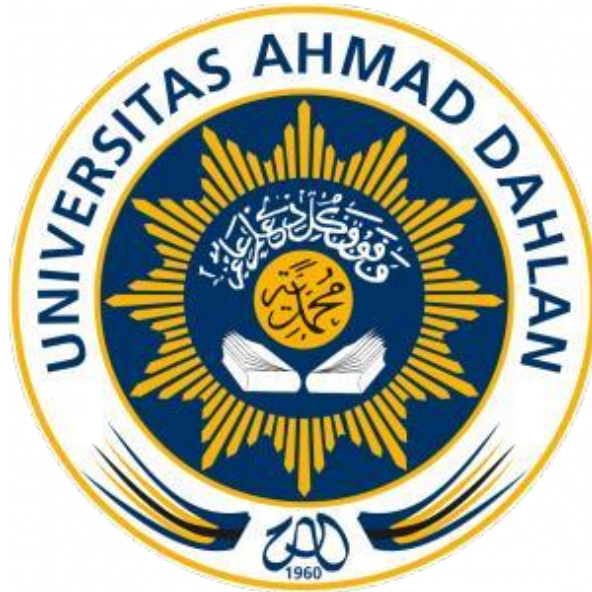


LAPORAN PRAKTIKUM VIII

KONSEP BASIS DATA



Disusun Oleh :

Muhammad Harits Arrozzaq (2400016087)

Kelas Praktikum / Semester :

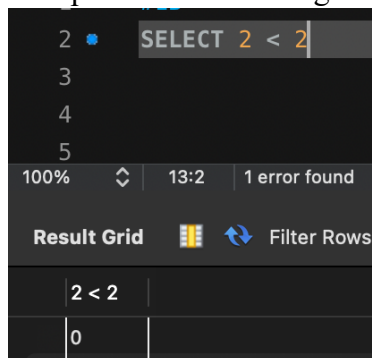
B/2

PROGRAM STUDI SISTEM INFORMASI
FAKULTAS SAINS DAN TEKNOLOGI TERAPAN
UNIVERSITAS AHMAD DAHLAN
YOGYAKARTA
2025

Tugas Praktikum:

1. Operator dan Fungsi Perbandingan

a. Operator Perbandingan.



2 • SELECT 2 < 2

3

4

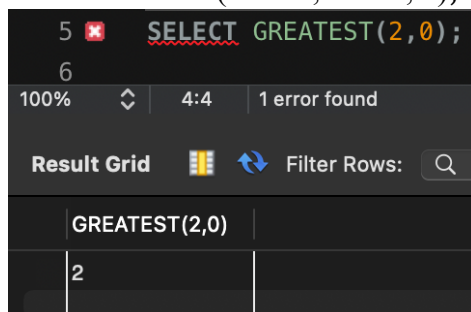
5

100% 13:2 1 error found

Result Grid Filter Rows

2 < 2
0

b. GREATEST(value1,value2,...), mengambil nilai terbesar.



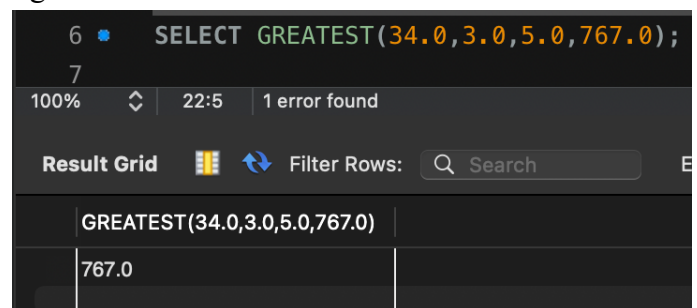
5 ✖ SELECT GREATEST(2,0);

6

100% 4:4 1 error found

Result Grid Filter Rows: Search

GREATEST(2,0)
2



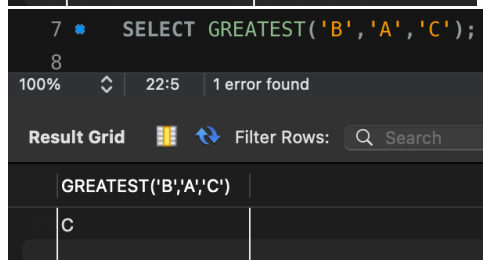
6 • SELECT GREATEST(34.0,3.0,5.0,767.0);

7

100% 22:5 1 error found

Result Grid Filter Rows: Search

GREATEST(34.0,3.0,5.0,767.0)
767.0



7 • SELECT GREATEST('B','A','C');

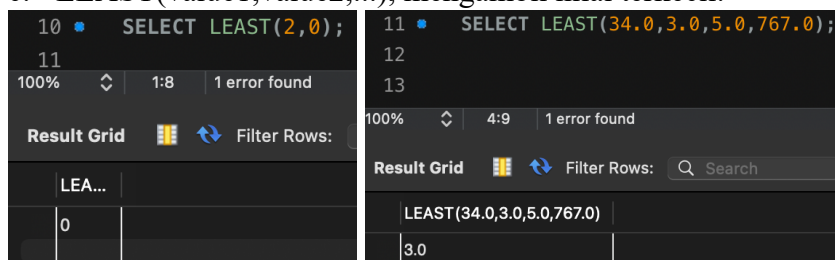
8

100% 22:5 1 error found

Result Grid Filter Rows: Search

GREATEST('B','A','C')
C

c. LEAST(value1,value2,...), mengambil nilai terkecil.



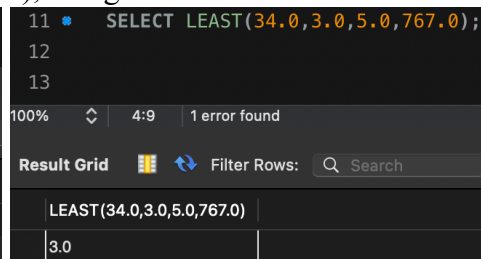
10 • SELECT LEAST(2,0);

11

100% 1:8 1 error found

Result Grid Filter Rows: Search

LEA...
0



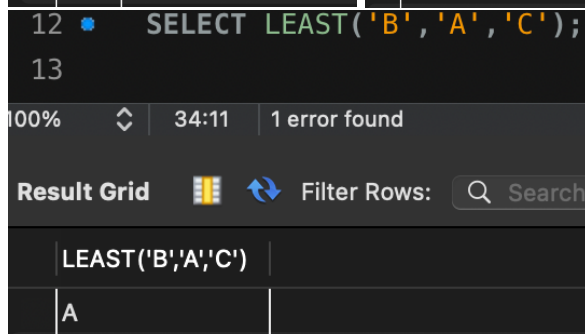
11 • SELECT LEAST(34.0,3.0,5.0,767.0);

12

100% 4:9 1 error found

Result Grid Filter Rows: Search

LEAST(34.0,3.0,5.0,767.0)
3.0



12 • SELECT LEAST('B','A','C');

13

100% 34:11 1 error found

Result Grid Filter Rows: Search

LEAST('B','A','C')
A

2. Operator Logika

a. NOT(!)

15 SELECT NOT 0;
16
100% 4:14 1 error found

Result Grid Filter Rows:

NOT 0
1

16 SELECT NOT NULL;
17
100% 14:15 1 error found

Result Grid Filter Rows:

NOT NULL
NULL

17 SELECT ! (1+1);
18
100% 17:16 1 error found

Result Grid Filter Rows:

! (1+1)
0

18 SELECT ! 1+1;
19
100% 16:17 1 error found

Result Grid Filter Rows:

! 1+1
1

b. AND (&&)

21 SELECT 1 && 1;
22
100% 4:20 1 error found

Result Grid Filter Rows:

1 && 1
1

22 SELECT 1 && 0;
23
100% 15:21 1 error found

Result Grid Filter Rows:

1 && 0
0

23 SELECT 1 && NULL;
24
100% 15:22 1 error found

Result Grid Filter Rows:

1 &...
NULL

24 SELECT 0 && NULL;
25
100% 18:23 1 error found

Result Grid Filter Rows:

0 &...
0

25 SELECT NULL && 0;
26
100% 18:24 1 error found

Result Grid Filter Rows:

NU...
0

c. OR (||)

28
SELECT 1 || 1;
29
100%
4:27
1 error found
Result Grid
Filter Rows:

1 1
1

29
SELECT 1 || 0;
30
100%
15:28
1 error found
Result Grid
Filter Rows:

1 0
1

30
SELECT 0 || 0;
31
100%
15:29
1 error found
Result Grid
Filter Rows:

0 0
0

31
SELECT 0 || NULL;
32
100%
15:30
1 error found
Result Grid
Filter Rows:

0...
NULL

32
SELECT 1 || NULL;
33
100%
15:30
1 error found
Result Grid
Filter Rows:

1...
1

d. XOR

35
SELECT 1 XOR 1;
36
100%
1:33
1 error found
Result Grid
Filter Rows:

1 XOR 1
0

36
SELECT 1 XOR 0;
37
100%
16:35
1 error found
Result Grid
Filter Rows:

1 XOR 0
1

37
SELECT 1 XOR NULL;
38
100%
16:36
1 error found
Result Grid
Filter Rows:

1 XOR NULL
NULL

38
SELECT 1 XOR 1 XOR 1;
39
100%
19:37
1 error found
Result Grid
Filter Rows:

1 XOR 1 XOR 1
1

3. Fungsi untuk kendali

a. IF(expr1, expr2, expr3)

41 • <code>SELECT IF(1>2,2,3);</code> 42	42 • <code>SELECT IF(1<2,'yes','no');</code> 43
100% 1:39 1 error found	100% 20:41 1 error found
Result Grid Filter Rows: Search	Result Grid Filter Rows: Search
IF(1>2,2,3)	IF(1<2,'yes','no')
3	yes

43 • <code>SELECT IF(STRCMP('test','test1'),'no','yes');</code> 44
100% 27:42 1 error found
Result Grid Filter Rows: Search Export: [icon]
IF(STRCMP('test','test1'),'no','yes')
no

b. IFNULL(expr1,expr2), jika expr1 tidak bernilai NULL, maka fungsi akan memberikan nilai expr1, sebaliknya expr2

46 • <code>SELECT IFNULL(1,0);</code> 47	47 • <code>SELECT IFNULL(NULL,10);</code> 48
100% 4:45 1 error found	100% 20:46 1 error found
Result Grid Filter Rows: Search	Result Grid Filter Rows: Search
IFNULL(1,0)	IFNULL(NULL,10)
1	10

48 • <code>SELECT IFNULL(1/0,10);</code> 49	49 • <code>SELECT IFNULL(1/0,'yes');</code> 50
100% 24:47 1 error found	100% 10:47 1 error found
Result Grid Filter Rows: Search	Result Grid Filter Rows: Search
IFNULL(1/0,10)	IFNULL(1/0,'yes')
10.0000	yes

c. NULLIF(expr1,expr2), jika expr1 = expr2, maka fungsi akan menghasilkan NULL, sebaliknya akan memberikan nilai expr1.

52 • <code>SELECT NULLIF(1,1);</code> 53	53 • <code>SELECT NULLIF(1,2);</code> 54
100% 1:50 1 error found	100% 4:51 1 error found
Result Grid Filter Rows: Search	Result Grid Filter Rows: Search
NULLIF(1,1)	NULLIF(1,2)
NULL	1

4. Fungsi untuk pengolahan string

- a. ASCII(str): menampilkan kode ASCII karakter pertama str

```
56 SELECT ASCII('2'); SELECT ASCII(2); SELECT ASCII('dx');
```

57
100% 1:54 1 error found

Result Grid Filter Rows: Search Export:

ASCII('dx')
100

- b. BIN(N) : menghasilkan nilai biner dari N

```
59 SELECT BIN(12);
```

60
100% 25:56 1 error found

Result Grid Filter Rows

BIN(12)
1100

- c. BIT_LENGTH(str) : menghasilkan panjang string str dalam satuan bit

```
62 SELECT BIT_LENGTH('text');
```

63
100% 16:59 1 error found

Result Grid Filter Rows: Search

BIT_LENGTH('text')
32

- d. CHAR(N,...) : menampilkan karakter dengan kode ASCII N

```
65 SELECT CAST(CHAR(77, 121, 83, 81, 76) AS CHAR(100));
```

66
100% 1:63 1 error found

Result Grid Filter Rows: Search Export:

CAST(CHAR(77, 121, 83, 81, 76) AS CHAR(100))
MySQL

```
66 SELECT CAST(CHAR(77, 77.3, '77.3') AS CHAR(100));
```

67
100% 53:65 1 error found

Result Grid Filter Rows: Search Export:

CAST(CHAR(77, 77.3, '77.3') AS CHAR(100))
MMM

- e. CHAR_LENGTH(str), LENGTH(str) : menghasilkan panjang string str dalam satuan karakter

- f. `CONCAT(str1,str2,...)` : menggabungkan string-string

70

SELECT CONCAT('My', 'S', 'QL');

71

00%

55:69

1 error found

Result Grid

Filter Rows:

Search

CONCAT('My', 'S', 'QL')
MySQL

72

SELECT CONCAT(14.3);

73

00%

33:71

1 error found

Result Grid

Filter Rows:

Search

CONCAT(14.3)
14.3

71

SELECT CONCAT('My', NULL, 'QL');

72

00%

32:70

1 error found

Result Grid

Filter Rows:

Search

CONCAT('My', NULL, 'QL')
NULL

- g. `CONCAT_WS(separator,str1,str2,...)` : menggabungkan string-string tetapi dengan karakter pemisah

<pre>75 • SELECT CONCAT_WS(',', 'Muhammad', 'Harits', 'Arrozzaq');</pre> <pre>76</pre> 100% 100:74 1 error found Result Grid Filter Rows: Search Export: <table><tr><th>CONCAT_WS(',', 'Muhammad', 'Harits', 'Arrozzaq')</th></tr><tr><td>Muhammad,Harits,Arrozzaq</td></tr></table>	CONCAT_WS(',', 'Muhammad', 'Harits', 'Arrozzaq')	Muhammad,Harits,Arrozzaq
CONCAT_WS(',', 'Muhammad', 'Harits', 'Arrozzaq')		
Muhammad,Harits,Arrozzaq		

- h. `CONV(N,from_base,to_base)` : mengkonversikan bilangan dari suatu basis ke basis lainnya

78



SELECT CONV('a',16,2);

79

100%



1:76

1 error found

Result Grid





Filter Rows:

 Search

CONV('a',16,2)
1010

79



SELECT CONV('6E',18,8);

80

100%



32:75

1 error found

Result Grid





Filter Rows:

 Search

CONV('6E',18,8)
172

- i. `LCASE(str)` , `LOWER(str)`.

<pre>82 • SELECT LOWER('QUADRATICALLY');</pre> <pre>83</pre> 100% 23:78 1 error found Result Grid Filter Rows: Search <table><tr><th>LOWER('QUADRATICALLY')</th></tr><tr><td>quadratically</td></tr></table>	LOWER('QUADRATICALLY')	quadratically
LOWER('QUADRATICALLY')		
quadratically		

- j. LEFT(str,len) : mengambil sejumlah karakter dari str pada sebelah kiri sejumlah len

85 • SELECT LEFT('foobarbar', 5);

86

100% 94:84 1 error found

Result Grid Filter Rows: Search

LEFT('foobarbar', 5)
fooba

- k. STRCMP(expr1,expr2) : membandingkan dua buah string, jika expr1=expr2 akan memberikan hasil 0, jika expr1 < expr2 hasilnya -1, sedang jika expr1 > expr2 hasilnya 1.

89 • SELECT STRCMP('text2', 'text');

90

100% 32:88 1 error found

Result Grid Filter Rows: Search

STRCMP('text2', 'text')
1

89 • SELECT STRCMP('text2', 'text');

Refresh data re-executing the original query

90

100% 32:88 1 error found

Result Grid Filter Rows: Search

STRCMP('text2', 'text')
1

90 • SELECT STRCMP('text', 'text');

91

100% 32:89 1 error found

Result Grid Filter Rows: Search

STRCMP('text', 'text')
0

5. Operator dan fungsi numerik

- a. Operator penjumlahan(+)

94 • SELECT 3+5;

95

100% 31:92 1 error found

Result Grid Filter Rows: Search

3+5
8

- b. Operator pengurangan(-)

96 • SELECT 3-5;

97

100% 26:95 1 error found

Result Grid Filter Rows: Search

3-5
-2

- c. Operator perkalian(*)

98 • SELECT 3*5;

99

100% 23:97 1 error found

Result Grid

	3*5
	15

- d. Operator pembagian(/)

100 • SELECT 3/5;

101

100% 25:99 1 error found

Result Grid

	3/5
	0.6000

6. Fungsi yang terkait dengan waktu (tanggal dan jam)

- a. Fungsi untuk menampilkan tanggal sekarang

103 • select current_date as Tgl_Sekarang;

104

100% 54:102 1 error found

Result Grid

	Tgl_Sekarang
	2025-06-19

- b. Fungsi untuk menampilkan waktu sekarang

107 • select current_time as Waktu_Sekarang;

108

100% 41:106 1 error found

Result Grid

	Waktu_Sekarang
	11:46:21

7. Fungsi agregasi (MIN, MAX, COUNT, AVG, dll, lihat pada DIKAT SQL)

- a. Fungsi MIN (menghasilkan nilai terkecil)

118 • select min(jumlah) from buku;

119

100% 1:115 1 error found

Result Grid

	min(jumlah)
	75

- b. Fungsi MAX (menghasilkan nilai terbesar)

```
121 • select max(jumlah) from buku;
```

122

100% 42:120 1 error found

Result Grid Filter Rows: Search

max(jumlah)
240

- c. Fungsi COUNT (menghitung cacah data)

```
124 • select count(nama_mahasiswa) from mahasiswa;
```

125

100% 38:123 1 error found

Result Grid Filter Rows: Search Export:

count(nama_mahasiswa)
5

- d. Fungsi AVG (menghitung nilai rata-rata)

```
127 • select avg(Jumlah) from buku;
```

128

100% 1:125 1 error found

Result Grid Filter Rows: Search

avg(Jumlah)
123.0000

- e. Fungsi SUM (menghitung jumlah data)

```
130 • select sum(jumlah) from buku;
```

131

100% 1:128 1 error found

Result Grid Filter Rows: Search

sum(jumlah)
615